

# Crypto Has Fundamentals:

## A Seven-Factor Model for Digital Asset Returns

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### Abstract

Cryptocurrency markets are widely perceived as speculative and disconnected from fundamentals. This perception keeps many traditional investors on the sidelines: without understanding what drives digital asset returns, they cannot decompose portfolio performance, hedge unwanted exposures, or distinguish skill from systematic risk. This paper addresses that gap. We introduce the QuantNest-7 (Q-7) model, a seven-factor framework that extends the three established factors of [Liu et al. \(2022\)](#) (market, size, and momentum) with four new dimensions drawn from blockchain data and derivatives markets: on-chain quality, fundamental value, residual volatility, and perpetual futures funding sentiment. Using

90 tokens over six years (January 2020 through May 2026), we find that five of six style factors carry statistically significant risk premia. On-chain quality, value, volatility, funding sentiment, and short-term reversal are all significant at the 1% level, while size becomes insignificant after controlling for volatility, suggesting the apparent size premium is a volatility premium in disguise. Notably, we document short-term return reversal rather than the momentum effect reported by [Liu et al. \(2022\)](#), indicating that past losers outperform past winners in our broader, more recent sample. The Q-7 model exhibits essentially zero overfitting, explains 63% of per-token return variation and 85% of diversified portfolio returns out of sample. Conditioning on market sentiment reveals regime-dependent premia: reversal earns its strongest returns during fearful markets, value during optimistic ones, while quality delivers positive returns across all sentiment environments. For practitioners, the Q-7 model provides the first comprehensive factor-based toolkit for digital assets: portfolio managers can decompose crypto returns into systematic exposures, construct factor-neutral portfolios, run performance attribution, and begin treating digital assets with the same analytical rigor applied to equities and fixed income.

**Keywords:** cryptocurrency, factor model, on-chain data, asset pricing, blockchain, sentiment regimes, short-term reversal

**JEL Classification:** G12, G14, G15

# 1 Introduction

What drives cryptocurrency returns? The answer matters for a growing number of institutional investors, pension funds, and family offices allocating to digital assets. Without a systematic factor model, portfolio managers cannot decompose performance, cannot hedge unwanted exposures, cannot benchmark against systematic risk factors, and cannot distinguish skill from factor tilts. Equities have had Fama-French and Barra for decades. Fixed income has term and credit factors. Digital assets, despite now representing a multi-trillion dollar market, lack an equivalent framework. This paper provides one.

The foundational work on this question belongs to [Liu et al. \(2022\)](#), who show that a three-factor model consisting of market, size, and momentum captures the cross-section of expected cryptocurrency returns. Their paper tests a comprehensive list of price-based characteristics and finds that the three-factor model subsumes all of them. It is an elegant result, but it leaves open an important question: what about the data that lives on the blockchain itself?

Unlike equities, where fundamental data arrives through quarterly earnings reports, blockchain networks produce a continuous stream of operational metrics. Every day, one can observe exactly how many unique addresses transacted on a network, how much economic value flowed through it, what fees users paid for block space, and how token supply is distributed across holders. These are direct, real-time measurements of network usage, far more granular than anything available for traditional securities. The question is whether any of it predicts returns.

[Sakkas and Urquhart \(2024\)](#) take a first pass at this question. They construct 13 factors from blockchain data and find that a parsimonious two-factor model (market plus a “network distribution” factor capturing whale concentration) explains cryptocurrency returns better than price-based alternatives. Their key insight is that decentralization matters: investors demand a premium for holding tokens with concentrated ownership.

We build on [Liu et al. \(2022\)](#) and [Sakkas and Urquhart \(2024\)](#). The QuantNest-7

model (hereafter Q-7) retains the market and size factors from [Liu et al. \(2022\)](#) and adds five new dimensions: short-term reversal, fundamental value, residual volatility, on-chain quality, and derivatives funding sentiment. Each captures a distinct economic mechanism through which token characteristics predict subsequent returns.

Four results stand out. First, on-chain quality dominates all other factors in cross-sectional predictive power, with a Newey–West  $t$ -statistic of 9.80. Tokens whose blockchain networks are genuinely used earn meaningfully higher subsequent returns. Second, we document short-term return reversal rather than momentum: at the 21-day horizon, past losers outperform past winners ( $t = 6.19$ ), contradicting the momentum effect reported by [Liu et al. \(2022\)](#). Third, after orthogonalizing volatility to size, the size premium becomes statistically insignificant ( $t = 1.36$ ), suggesting that the apparent size effect in cryptocurrency markets is largely a volatility premium in disguise. Fourth, the funding rate signal is strongly priced ( $t = 6.74$ ) and nearly orthogonal to all other factors, capturing a sentiment dimension that no existing model includes.

The Q-7 model exhibits essentially zero overfitting, with an in-sample/out-of-sample gap of  $-0.001$ . A diversified portfolio of 30 tokens achieves 85% out-of-sample  $R^2$ , indicating that the model captures the dominant systematic return drivers relevant for real-world portfolio management.

The paper proceeds as follows. Section 2 describes the data. Section 3 defines each factor conceptually. Section 4 presents the empirical findings. Section 5 evaluates the model in and out of sample. Section 6 offers robustness checks. Section 7 discusses limitations, and Section 8 concludes.

## 2 Data

### 2.1 Sources

We draw exclusively on Coin Metrics, an institutional-grade provider of blockchain network data and reference rates. Coin Metrics constructs daily reference prices using volume-weighted median, time-weighted average, and inverse price variance-

weighted techniques across vetted exchanges. This methodology addresses the data quality concerns associated with single-exchange prices or self-reported volume figures that affect much of the existing cryptocurrency literature.

For 53 of our 90 tokens, Coin Metrics provides daily on-chain metrics including active address counts, transaction volumes, fee revenue, and derived valuation ratios. For tokens where Coin Metrics tracks blockchain data under chain-specific identifiers (e.g., Avalanche C-Chain activity or SHIB activity on the Ethereum network), we merge the on-chain metrics with the base token’s price and market capitalization. This recovers on-chain coverage for 18 additional tokens. Perpetual futures funding rates are sourced through Coin Metrics’ market data feed, covering 89 tokens across major derivatives exchanges.

The decision to rely on a single data provider is deliberate. Multi-source approaches introduce reconciliation challenges (different pricing methodologies, inconsistent timestamps, varying quality standards) that are difficult to control in an academic setting. Coin Metrics’ institutional client base and transparent methodology provide a level of data integrity that simplifies interpretation of our results.

**Table 1:** Data Sources

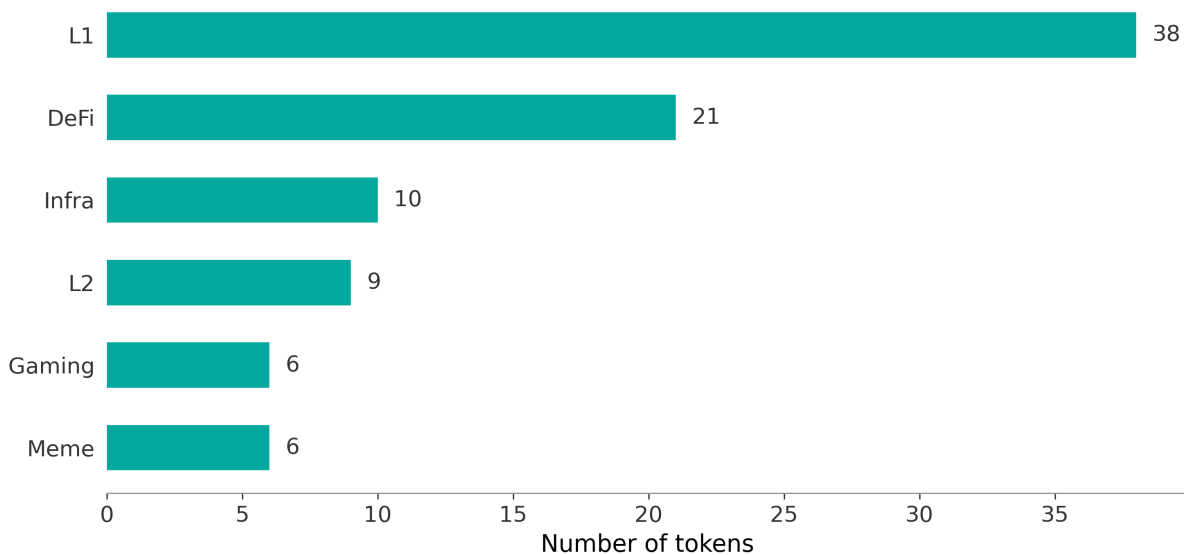
| Source           | Data Type                          | Assets | Frequency |
|------------------|------------------------------------|--------|-----------|
| Coin Metrics Pro | On-chain metrics, reference prices | 53     | Daily     |
| Coin Metrics Pro | Market capitalization              | 90     | Daily     |
| Coin Metrics Pro | Perpetual funding rates            | 89     | 8-hourly  |

## 2.2 Universe

Tokens enter the universe if they have daily price and market capitalization data available from Coin Metrics. We exclude stablecoins, wrapped tokens, tokenized commodities, and exchange-native tokens. The surviving universe contains 90 unique tokens assigned to six broad sectors: Layer-1 blockchain platforms (38 tokens), decentralized finance protocols (21), infrastructure tokens (10), Layer-2 scaling solutions (9), meme tokens (6), and gaming tokens (6).

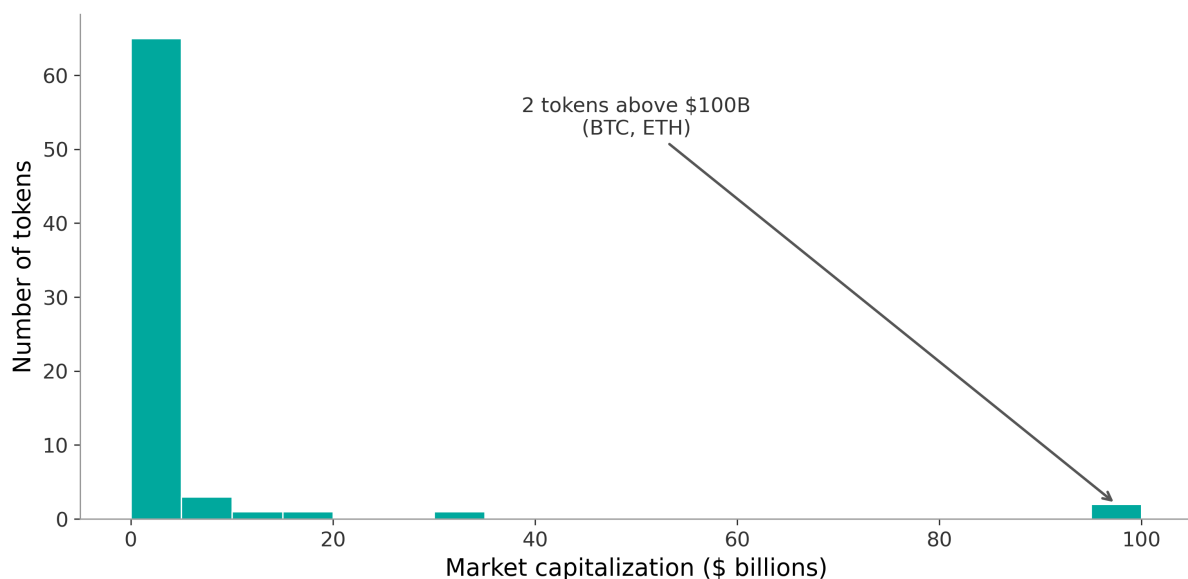
The sample runs from January 2, 2020 through May 13, 2026, totaling 2,324 trading days. Token coverage grows from 33 in early 2020 to 90 by late 2025 as newer projects enter the universe. The final panel contains 170,006 token-day observations, with an average of 73 tokens per day.

One important caveat: our universe is constructed from tokens that exist and trade actively at the end of the sample period, then projected backward. Tokens that collapsed (LUNA, FTT) are absent. This survivorship bias systematically inflates all factor returns and is a limitation shared with nearly all existing cryptocurrency factor research.



**Figure 1:** Token count by sector.

As Figure 1 shows, Layer-1 platforms dominate the universe with 38 tokens. These include both smart contract networks such as Ethereum, Solana, and Avalanche, and payment-oriented chains such as Bitcoin, Litecoin, and XRP. DeFi (21 tokens) encompasses decentralized exchanges, lending protocols, and liquid staking providers. Infrastructure (10 tokens) includes oracle networks, storage providers, and compute marketplaces. Layer-2 (9 tokens) covers scaling solutions built on top of existing blockchains. Gaming (6 tokens) includes metaverse and play-to-earn platforms. Meme tokens (6) are community-driven tokens with no explicit utility beyond speculation.



**Figure 2:** Market capitalization distribution across the 90-token universe.

As Figure 2 illustrates, the distribution is heavily right-skewed: the majority of tokens have market capitalizations below \$10 billion, while a small number of large-cap tokens (Bitcoin, Ethereum, BNB, Solana, XRP) occupy the upper tail. Values above \$100 billion are clipped for readability.

## 2.3 Summary Statistics

Table 2 characterizes the panel. Daily returns average 0.15% with a standard deviation of 6.0%, confirming the high volatility that defines this asset class. The average cross-sectional return dispersion of 4.74% per day is roughly 2.5 times the typical daily cross-sectional dispersion observed in the S&P 500 (Fama and French, 1993), providing substantial scope for cross-sectional factor strategies to differentiate between outperformers and underperformers on any given day.

**Table 2:** Panel Descriptive Statistics

|                                    | Mean | Std Dev | Skewness | Min   | Max   |
|------------------------------------|------|---------|----------|-------|-------|
| Daily return (%)                   | 0.15 | 6.0     | 1.26     | -78.0 | 79.0  |
| Market cap (\$B)                   | 22.7 | 138.7   | 10.94    | 0.0   | 2,488 |
| Cross-sectional dispersion (%/day) | 4.74 | —       | —        | —     | —     |
| Tokens per day                     | 73   | —       | —        | 33    | 90    |

### 3 Factor Definitions

The Q-7 model consists of seven factors. All factor exposures are computed cross-sectionally each day and standardized to zero mean and unit variance. Factor returns are the equal-weighted mean return of the top quintile minus the bottom quintile. To ensure that factors capture independent sources of return variation, we apply a selective orthogonalization procedure. Rather than orthogonalizing each factor to all others, we remove only the overlaps where there is a clear economic rationale. For each orthogonalized factor, we regress its daily cross-sectional exposures on the control factors listed below and retain only the residual:

1. **Market:** capitalization-weighted return. No orthogonalization.
2. **Size:** negative log market capitalization. No orthogonalization. Used as a control for other factors because many token characteristics (volatility, valuation, activity) correlate mechanically with market capitalization.
3. **Reversal:** negated 21-day return. No orthogonalization.
4. **Value:** on-chain valuation composite, orthogonalized to Size. Rationale: smaller tokens tend to appear mechanically cheaper on certain valuation ratios. Removing the size component isolates genuine fundamental cheapness.
5. **Volatility:** residual volatility, orthogonalized to Size. Rationale: smaller tokens are mechanically more volatile. Without this step, the volatility factor would partly proxy for size.
6. **Quality:** on-chain activity composite, orthogonalized to Size and Value. Rationale: larger networks tend to have more absolute activity (size overlap), and fundamentally cheap tokens may coincide with high-activity ones (value overlap).
7. **Funding:** perpetual funding rate, orthogonalized to Size and Reversal. Rationale: smaller tokens may have systematically different funding dynamics (size

overlap), and recent losers may attract contrarian funding flows (reversal overlap).

An important distinction: this orthogonalization operates on daily cross-sectional exposures, not on the time-series of factor returns. Two factors can have uncorrelated exposures on any given day yet still exhibit correlated returns over time, because the specific tokens in each quintile portfolio shift from day to day and market-wide moves affect different long-short portfolios in correlated ways. This is why the factor return correlation table (Table 5) shows non-zero correlations despite the orthogonalization.

All portfolio weights and factor inputs use beginning-of-period (lagged) market capitalization to prevent look-ahead bias.

### **3.1 Market (MKT)**

The market factor is the capitalization-weighted return of all tokens in the universe on each day, with weights based on lagged market capitalization. A minimum of 10 tokens is required for the market return to be computed. The market factor captures the common component of cryptocurrency returns and serves as the dominant systematic risk exposure in the model.

### **3.2 Size (SMB)**

The size factor ranks tokens each day by the negative logarithm of their lagged market capitalization, so that smaller tokens receive positive exposure. The economic rationale is that small-capitalization tokens may carry a liquidity premium, a higher sensitivity to market-wide shocks, or greater information asymmetry. However, as we document in the results, the size premium becomes statistically insignificant after controlling for volatility, suggesting that any apparent size effect operates primarily through the volatility channel. Size is retained in the model because it serves as an orthogonalization control for the value, quality, and funding factors.

### 3.3 Short-Term Reversal (REV)

We compute the 21-day cumulative return for each token with a one-day skip period (to avoid microstructure-driven noise on the most recent day), following the methodology of [Jegadeesh and Titman \(1993\)](#). We then negate the signal, so that tokens with the worst recent performance receive positive exposure. This captures the well-documented tendency for short-term losers to outperform in subsequent periods, a pattern attributed to overreaction, liquidity-driven selling pressure, and mean reversion in speculative markets.

This is a notable departure from [Liu et al. \(2022\)](#), who document a momentum (continuation) effect. In our broader 90-token sample with more recent data (extending through 2026), past losers consistently outperform past winners across all lookback horizons tested. We discuss this finding further in the robustness section.

### 3.4 Value (VAL)

Blockchain networks generate daily observable measures of economic activity, and the ratio of a token’s market valuation to these activity measures provides a natural notion of “cheapness.” Our value factor aggregates five proprietary on-chain valuation ratios into a single composite score. Each individual ratio compares a token’s market capitalization to a different measure of its underlying network economic output. The composite uses a rank-based aggregation methodology that is robust to outliers and handles varying data coverage across tokens. Tokens that appear undervalued relative to their on-chain fundamentals receive positive exposure. The value factor is orthogonalized to size to ensure that it captures genuine fundamental cheapness rather than the mechanical tendency for smaller tokens to appear cheaper on certain metrics.

### 3.5 Volatility (VOL)

The volatility factor measures idiosyncratic risk after removing systematic market exposure. For each token, we estimate a time-varying beta relative to the Q-7 market

factor using an exponentially weighted methodology that places greater emphasis on recent observations. We then compute the residual return (the token’s actual return minus its beta-implied market return) and measure the exponentially weighted standard deviation of these residuals.

Tokens are sorted by their residual volatility, and the signal is negated so that low-volatility tokens receive positive exposure. This construction is motivated by the low-volatility anomaly literature in equities ([Baker et al., 2011](#)), where low-volatility stocks have historically outperformed on a risk-adjusted basis. As we document below, this relationship is inverted in cryptocurrency markets: high-volatility tokens outperform, consistent with a genuine risk premium for bearing idiosyncratic risk. The volatility factor is orthogonalized to size to separate the volatility premium from any mechanical correlation with market capitalization.

### 3.6 Quality (QUA)

The quality factor captures whether a blockchain network is genuinely being used. We construct a proprietary composite from four daily on-chain activity indicators sourced from Coin Metrics. These indicators measure, through different lenses, the breadth and intensity of genuine network activity: the number of unique participants engaging with the network, the economic value flowing through it, the transaction throughput, and the fees that users willingly pay for block space. Each signal is transformed to reduce the influence of outliers and ranked cross-sectionally before averaging into a composite score.

The quality composite is orthogonalized to both size and value. This ensures that the quality signal captures the independent contribution of network activity, stripped of any mechanical correlation with how large or cheap a token happens to be. Tokens with high on-chain usage receive positive exposure.

Quality is available for approximately 53 tokens (those with sufficient on-chain data from Coin Metrics), resulting in quintiles of about 10 tokens each. Despite this relatively narrow cross-section, the statistical significance of the results is striking, as

we document below.

### 3.7 Funding Sentiment (FND)

Perpetual futures are the dominant derivative instrument in cryptocurrency markets. Unlike standard futures contracts, perpetuals have no expiry date. Instead, they use a funding mechanism to keep the futures price tethered to the spot price: periodically (typically every eight hours), if the futures price is above spot, long holders pay short holders a “funding rate,” and vice versa. When funding rates are persistently positive, it signals that speculative long positions are crowded and that longs are paying a premium to maintain their exposure.

We use this as a contrarian indicator. Tokens with elevated funding rates (indicating excessive bullish crowding) receive negative exposure, on the premise that crowded trades tend to unwind. The funding factor is orthogonalized to size and reversal. Funding data is available for 89 of 90 tokens in the universe.

### 3.8 Sector Controls

In addition to the seven style factors, we include six sector dummy variables as control variables in the Fama–MacBeth regressions. The six sectors are Layer-1, Layer-2, DeFi, Infrastructure, Meme, and Gaming, with a residual “Other” category serving as the reference group.

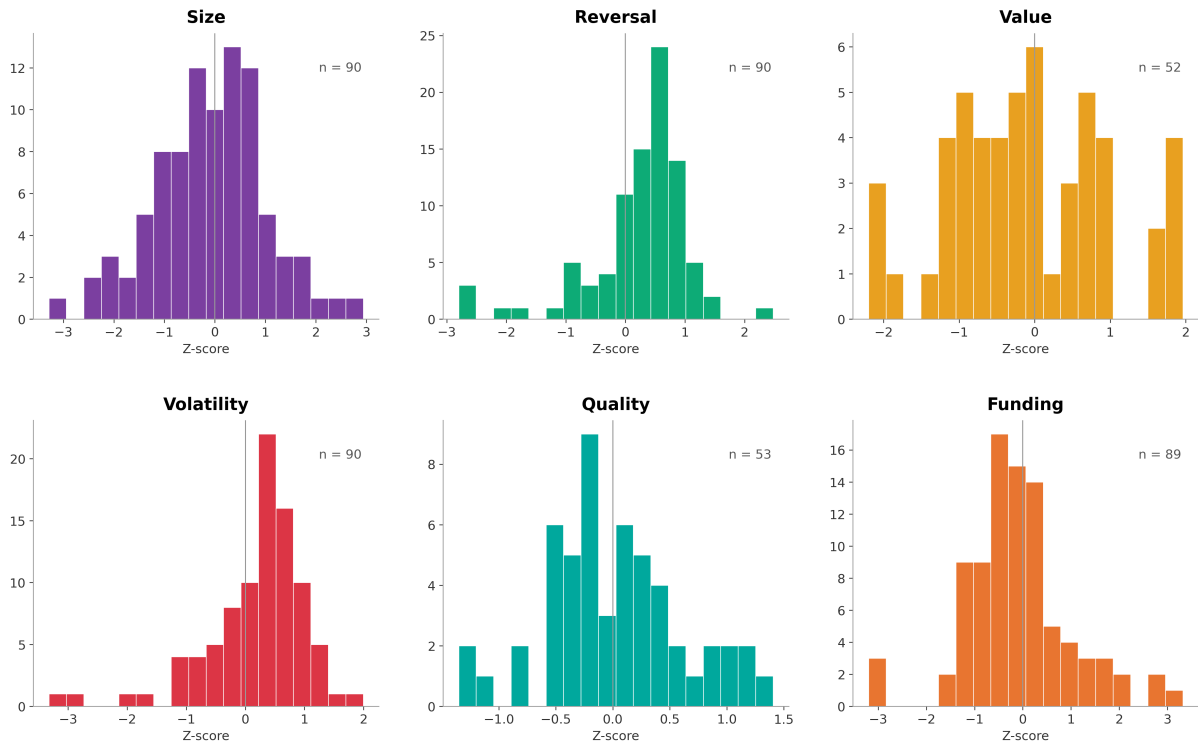
Sector controls are not style factors. A style factor (like value or quality) represents a continuous characteristic that tokens possess to varying degrees, allowing a meaningful ranking from low to high. Sector membership, by contrast, is binary: a token either belongs to DeFi or it does not. There is no “top quintile of DeFi-ness.” Including sector dummies as controls serves a different purpose: it absorbs return variation attributable to broad sector trends (e.g., a DeFi rally lifting all DeFi tokens) so that the style factor coefficients capture genuine token-level, within-sector effects. Without sector controls, a positive quality coefficient could partly reflect the fact that high-quality tokens tend to cluster in sectors that happened to outperform during the sample.

### 3.9 Coverage

Not all factors are available for all tokens. Table 3 summarizes the coverage. Price-based factors (size, reversal, volatility) cover the full 90-token universe. On-chain factors (value, quality) require blockchain data and are available for 52 to 53 tokens, representing approximately 90% of total crypto market capitalization. Funding requires a listed perpetual contract and covers 89 tokens. In Fama–MacBeth regressions, tokens without a particular factor exposure receive a value of zero, effectively assigning them neutral exposure to that factor.

**Table 3: Factor Coverage**

| Factor           | Coverage (%) | Tokens |
|------------------|--------------|--------|
| MKT (Market)     | 100.0        | 90     |
| SMB (Size)       | 98.1         | 90     |
| REV (Reversal)   | 98.8         | 90     |
| VAL (Value)      | 60.6         | 52     |
| VOL (Volatility) | 97.7         | 90     |
| QUA (Quality)    | 62.1         | 53     |
| FND (Funding)    | 90.6         | 89     |



**Figure 3: Cross-sectional distribution of factor exposures on the final sample date.**

Figure 3 confirms that size and reversal cover the full universe, producing well-populated distributions centered near zero. Value and quality have narrower coverage (52 and 53 tokens, respectively) due to on-chain data requirements. All distributions are approximately symmetric, confirming proper z-score normalization.

## 4 Results

### 4.1 Factor Return Characteristics

Table 4 reports annualized statistics for each factor’s long-short quintile portfolio over the full sample period. Five of the six style factors earn positive annualized returns: quality, value, reversal, funding, and size. The volatility factor earns a large negative return ( $-234\%$  annualized), indicating that high-volatility tokens outperform low-volatility tokens throughout the sample. This is the opposite of the low-volatility anomaly documented in equities (Baker et al., 2011) and reflects a genuine feature of cryptocurrency markets, where the marginal investor is a speculative retail participant who is compensated for bearing idiosyncratic risk rather than penalized for it.

Quality and value exhibit the most attractive return profiles, combining high annualized returns with moderate volatility and relatively shallow maximum drawdowns ( $-42\%$  and  $-24\%$ , respectively). The reversal factor produces the highest annualized return ( $+151\%$ ) but also the highest kurtosis (22.4), indicating that its returns are driven by occasional extreme moves rather than steady accumulation.

We caution against reading these as achievable strategy returns. Survivorship bias, transaction costs, and limited capacity in smaller tokens would substantially reduce real-world implementation returns.

**Table 4:** Factor Return Statistics (Annualized)

| Factor | Return  | Volatility | Max DD  | Skew  | Kurt |
|--------|---------|------------|---------|-------|------|
| MKT    | 56.6%   | 63.4%      | −77.3%  | −0.80 | 10.7 |
| SMB    | 25.0%   | 54.9%      | −86.1%  | −0.38 | 7.5  |
| REV    | 150.5%  | 70.7%      | −82.0%  | 1.86  | 22.4 |
| VAL    | 184.5%  | 45.5%      | −23.7%  | 0.71  | 3.9  |
| VOL    | −234.4% | 96.3%      | −100.0% | 0.24  | 4.8  |
| QUA    | 244.5%  | 53.7%      | −42.2%  | 0.45  | 2.2  |
| FND    | 135.4%  | 44.8%      | −78.3%  | 0.78  | 3.2  |

**Sample:** Annualized using a 365-day convention. Factor returns are equal-weighted top-minus-bottom quintile spreads. Sample: January 2020 to May 2026, 2,324 days, 90 tokens.

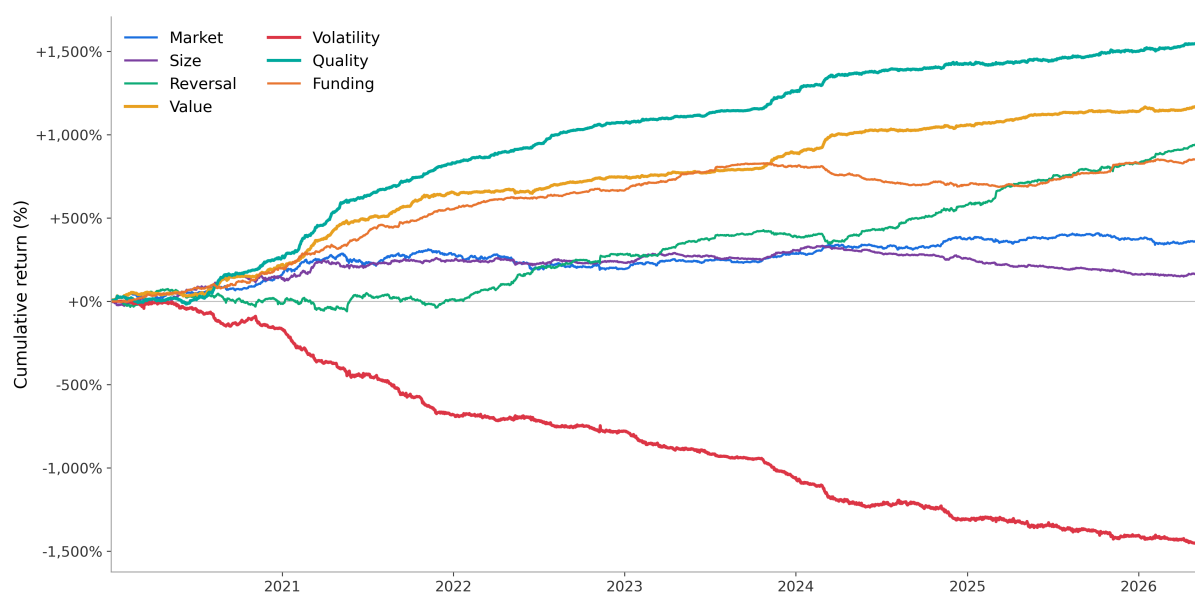
**Figure 4:** Cumulative factor returns over the full sample period.

Figure 4 reveals several patterns. Quality accumulates the most smoothly, with its cumulative return growing steadily through both bull and bear markets. Volatility trends persistently downward, confirming the inverted volatility premium in crypto. The value factor is flat through 2020–2021 and then accelerates sharply from 2023 onward, coinciding with the market’s shift toward fundamentals-driven investing.

## 4.2 Factor Independence

Table 5 reports pairwise correlations of daily factor returns. These are return correlations, not exposure correlations. As discussed in Section 3, our orthogonalization procedure ensures that factor exposures are uncorrelated in the daily cross-section, but the resulting long-short portfolio returns can still co-move over time. The largest absolute return correlation is  $-0.56$  between size and volatility, reflecting the persistent tendency for small tokens and high-volatility tokens to move together in response to market-wide risk-on/risk-off dynamics. The value-quality correlation of  $+0.35$  arises because both factors draw on overlapping on-chain data for a similar subset of tokens. The funding factor is nearly uncorrelated with everything else (maximum  $|\rho| = 0.17$ ), confirming that derivatives sentiment captures a return dimension that price-based and on-chain factors miss entirely.

**Table 5:** Factor Return Correlations

|     | MKT     | SMB     | REV     | VAL     | VOL     | QUA  | FND  |
|-----|---------|---------|---------|---------|---------|------|------|
| MKT | 1.00    |         |         |         |         |      |      |
| SMB | $-0.40$ | 1.00    |         |         |         |      |      |
| REV | 0.10    | $-0.15$ | 1.00    |         |         |      |      |
| VAL | 0.08    | 0.04    | $-0.10$ | 1.00    |         |      |      |
| VOL | 0.43    | $-0.56$ | 0.15    | $-0.16$ | 1.00    |      |      |
| QUA | $-0.07$ | 0.16    | $-0.05$ | 0.35    | $-0.41$ | 1.00 |      |
| FND | $-0.00$ | 0.06    | 0.05    | 0.07    | $-0.17$ | 0.12 | 1.00 |

## 4.3 Quintile Monotonicity

A well-constructed factor should exhibit a monotonic pattern across quintile portfolios: returns should increase (or decrease) systematically from the bottom to the top quintile. Monotonicity provides evidence that the factor captures a continuous gradient in expected returns rather than a statistical artifact driven by extreme observations in a single quintile.

Table 6 reports annualized returns by quintile for each factor. Volatility displays the cleanest monotonic pattern, with returns declining steadily from  $+223\%$  in the lowest-volatility quintile to  $-12\%$  in the highest. Quality shows strong directionality,

with the bottom two quintiles earning negative returns ( $-58\%$  and  $-30\%$ ) and the top three earning positive returns, though the top quintile ( $+187\%$ ) is slightly below the fourth ( $+190\%$ ). Value and funding show correct directionality at the extremes (the top quintile substantially outperforms the bottom) with some noise in the intermediate quintiles, which is expected given the relatively narrow cross-sections involved. The reversal factor shows a steep gradient between Q1 ( $+57\%$ ) and Q5 ( $+207\%$ ), though Q2 is anomalously negative.

**Table 6:** Annualized Returns by Quintile (%)

|     | Q1 (Low) | Q2  | Q3  | Q4   | Q5 (High) | Monotonic? |
|-----|----------|-----|-----|------|-----------|------------|
| SMB | +66      | +57 | +72 | +80  | +91       | Approx.    |
| REV | +57      | -26 | +20 | +96  | +207      | Approx.    |
| VAL | -28      | -4  | +41 | +192 | +157      | Approx.    |
| VOL | +223     | +56 | +42 | +1   | -12       | Yes        |
| QUA | -58      | -30 | +71 | +190 | +187      | Approx.    |
| FND | +33      | +13 | +9  | +56  | +169      | Approx.    |

## 5 Model Evaluation

We split the sample at June 2024 (approximately 70/30), giving 1,626 in-sample and 698 out-of-sample days. The in-sample period includes the 2020–2021 bull run, the 2022 crash, and the recovery through mid-2024. The out-of-sample period covers the subsequent market through May 2026. We evaluate the model using three complementary approaches.

### 5.1 Cross-Sectional Explanatory Power

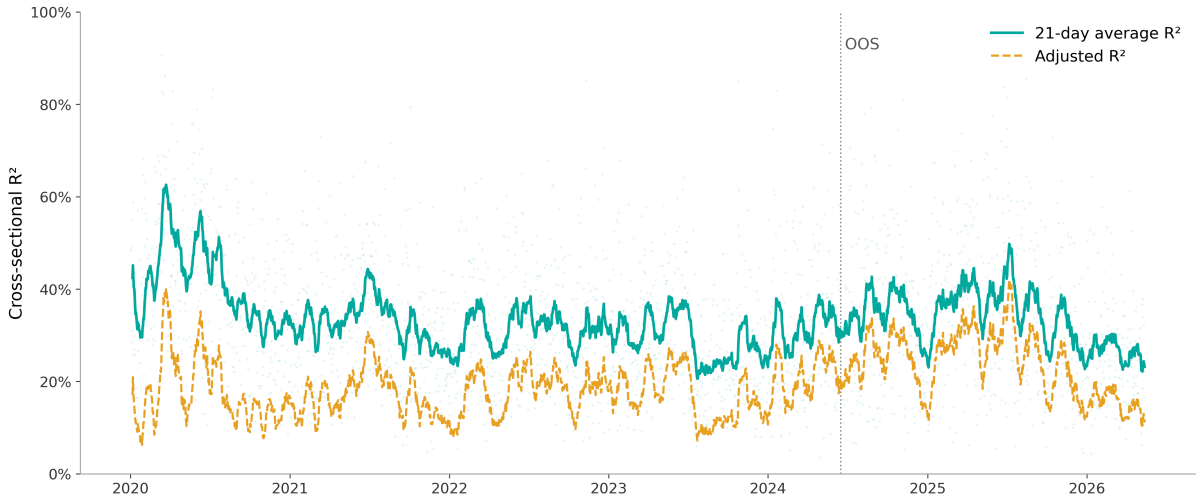
Each day, we run a Fama–MacBeth regression (Fama and MacBeth, 1973) of token returns on their lagged factor exposures. Table 7 reports the time-series average of daily  $R^2$  values. The Q-7 style factors alone explain 23% of daily cross-sectional return variation, rising to 33% with sector controls. These numbers are stable across sample periods: the out-of-sample  $R^2$  (0.335) is virtually identical to the in-sample value (0.334), producing an IS/OOS gap of  $-0.001$ . This level of stability is unusual in factor

model research and suggests that the factor structure is genuine rather than an artifact of in-sample optimization.

The relatively modest  $R^2$  levels (compared to, say, equity factor models applied to industry portfolios) reflect the inherently noisy nature of daily cryptocurrency returns. With an average of 73 tokens per cross-section, a substantial fraction of daily return variation is token-specific and unpredictable by any systematic model.

**Table 7:** Cross-Sectional  $R^2$  (Fama–MacBeth)

| Specification      | In-Sample | Out-of-Sample |
|--------------------|-----------|---------------|
| Style factors only | 0.231     | 0.274         |
| Adjusted           | 0.153     | 0.223         |
| Style + Sectors    | 0.334     | 0.335         |
| Adjusted           | 0.178     | 0.234         |



**Figure 5:** Daily Fama–MacBeth  $R^2$  with 21-day moving average and adjusted  $R^2$ .

## 5.2 Factor Risk Premia

Table 8 reports the central question of the paper: does exposure to each factor earn a statistically significant risk premium? We compute  $t$ -statistics using Newey–West standard errors with five lags to account for autocorrelation in the daily coefficient estimates.

Five of six style factors are significant at the 1% level. Quality leads with  $t = 9.80$ , indicating that on-chain network activity is the most reliable predictor of cross-

sectional return differences. Volatility follows with  $t = -9.19$  (negative because low-volatility tokens receive positive exposure, yet high-volatility tokens outperform). Value ( $t = 8.91$ ) confirms that fundamental cheapness relative to on-chain activity predicts higher returns. Funding ( $t = 6.74$ ) validates the contrarian crowding signal. Reversal ( $t = 6.19$ ) confirms that past losers outperform.

The notable exception is size. After orthogonalizing volatility to size, the size coefficient drops to an insignificant  $t = 1.36$ . This result has a clean economic interpretation: in cryptocurrency markets, small tokens tend to be more volatile. What previous studies identified as a “size premium” appears to be compensation for bearing idiosyncratic volatility rather than a genuine size effect. Once the volatility factor absorbs the volatility component that was mechanically correlated with size, the residual size effect disappears.

**Table 8:** Fama–MacBeth Risk Premia (In-Sample)

| Factor | Avg. Coeff. | $t_{\text{NW}}$ | $t_{\text{IID}}$ |     |
|--------|-------------|-----------------|------------------|-----|
| SMB    | +0.000306   | +1.36           | +1.17            |     |
| REV    | +0.002372   | +6.19           | +6.37            | *** |
| VAL    | +0.002288   | +8.91           | +10.16           | *** |
| VOL    | −0.004572   | −9.19           | −7.46            | *** |
| QUA    | +0.003770   | +9.80           | +10.44           | *** |
| FND    | +0.001808   | +6.74           | +6.59            | *** |

$t_{\text{NW}}$ : Newey–West with 5 lags.

$t_{\text{IID}}$ : no autocorrelation adjustment.

\*\*\* denotes significance at 1%. Average tokens per regression: IS=66, OOS=90.

### 5.3 Time-Series and Portfolio Fit

For each individual token, we regress its daily returns on the Q-7 factor return series to measure how much of that token’s return variation the model captures. The mean per-token  $R^2$  is 63% out of sample when using both style and sector factors, indicating that the model explains nearly two-thirds of individual token return variation on average. Major tokens such as Bitcoin and Ethereum achieve substantially higher  $R^2$  values.

For practitioners, the more relevant test is whether the model explains diversi-

fied portfolios. We construct 200 random equal-weighted portfolios at each size and regress their returns on the Q-7 factors. As Table 9 shows, diversification rapidly eliminates idiosyncratic noise: at 30 tokens, the model captures 85% of portfolio return variation out of sample.

**Table 9:** Time-Series and Portfolio  $R^2$

|                             | <b>In-Sample</b> | <b>Out-of-Sample</b> |
|-----------------------------|------------------|----------------------|
| Per-token (style only)      | 0.446            | 0.580                |
| Per-token (style + sectors) | 0.533            | 0.632                |
| Tokens analyzed             | 86               | 90                   |
| Portfolio: 5 tokens         | 0.670            | 0.760                |
| Portfolio: 10 tokens        | 0.759            | 0.803                |
| Portfolio: 20 tokens        | 0.825            | 0.835                |
| Portfolio: 30 tokens        | 0.851            | 0.848                |

## 5.4 Incremental Factor Contributions

To assess each factor’s independent contribution, we drop one factor at a time from the full model and measure the decline in mean out-of-sample  $R^2$ . The market factor is by far the largest contributor (23.4 percentage points), confirming that cryptocurrency returns remain dominated by a common market factor. Among style factors, volatility contributes the most (2.9 pp), followed by size (1.4 pp). Despite its strong  $t$ -statistic, quality’s incremental  $R^2$  contribution (0.4 pp) is modest because it is available for only 53 of 90 tokens, limiting its aggregate impact on the cross-section. The near-zero individual contributions of reversal, value, and funding do not imply these factors are unimportant; rather, their explanatory power overlaps with the other style factors when measured through the lens of time-series  $R^2$ .

**Table 10:** Incremental Factor Contributions (OOS Leave-One-Out)

| Dropped    | $\Delta R^2$ | % of Total |
|------------|--------------|------------|
| Market     | +0.234       | 37.0       |
| Volatility | +0.029       | 4.6        |
| Size       | +0.014       | 2.2        |
| Quality    | +0.004       | 0.7        |
| Funding    | +0.003       | 0.5        |
| Value      | +0.003       | 0.5        |
| Reversal   | +0.002       | 0.4        |

Full model OOS mean  $R^2 = 0.632$  across 90 tokens.  $\Delta R^2$  is the decrease in mean  $R^2$  when the named factor is excluded.

## 6 Robustness

### 6.1 Reversal Across Lookback Horizons

A natural concern is that the reversal finding is an artifact of the specific 21-day lookback window. Table 11 tests five alternative specifications. The reversal effect is present at all horizons, with shorter lookbacks producing the strongest returns. Performance is strongest at 7 days (+266% annualized) and weakens as the horizon extends, but remains positive even at 63 days (+116%). This pattern is consistent with the literature on short-term overreaction and mean reversion in speculative markets: prices overshoot in the short term and correct over the following weeks.

**Table 11:** Reversal Lookback Sensitivity

| Lookback | Ann. Return | Ann. Vol. | Max DD  |
|----------|-------------|-----------|---------|
| 7 days   | +266.1%     | 81.5%     | −100.0% |
| 14 days  | +200.1%     | 74.6%     | −100.0% |
| 21 days* | +151.3%     | 71.0%     | −82.0%  |
| 42 days  | +109.5%     | 65.9%     | −100.0% |
| 63 days  | +115.7%     | 63.7%     | −100.0% |

\* Q-7 model specification.

## 6.2 Sector-Neutral Reversal

One potential explanation for the reversal effect is that it simply captures sector rotation: if DeFi crashed last month and is now rebounding, a cross-sectional reversal signal would mechanically buy DeFi tokens and sell whatever rallied. Under this interpretation, the signal would have little to do with individual token characteristics and everything to do with sector-level mean reversion.

To test this, we construct a sector-neutral version of the reversal factor. Instead of ranking tokens against the full cross-section, we first standardize returns within each sector, then rank. This strips out any sector-level return pattern and isolates within-sector, token-level mean reversion.

The sector-neutral specification produces an annualized return of +202%, stronger than the raw cross-sectional version (+151%). Rather than weakening the signal, removing sector effects actually strengthens it. This confirms that the reversal effect is driven by individual tokens overshooting and correcting within their own sector, not by sector-level rotation.

## 6.3 Sentiment Regimes and Factor Premia

We merge our factor return series with the **Alternative.me Crypto Fear & Greed Index**, a daily composite scored from 0 (extreme fear) to 100 (extreme greed). Using the 30-day smoothed index, we classify each trading day into **Fear** (below 35, 746 days or 32% of the sample), **Neutral** (35 to 65, 947 days or 41%), or **Greed** (above 65, 613 days or 27%).

Table 12 reveals that factor premia vary substantially across sentiment regimes. The reversal factor performs strongly during fear (annualized return +235%) but is essentially flat during greed (−5%). This is economically intuitive: during market stress, panic selling drives prices below fundamental values, and the subsequent bounce-back rewards investors who buy recent losers. In euphoric markets, momentum-like continuation is more common, and the reversal signal weakens.

Value shows the opposite pattern, earning +94% during fear but +365% during

greed. In optimistic markets, capital flows toward tokens with demonstrable on-chain economic activity relative to their valuation, widening the value spread.

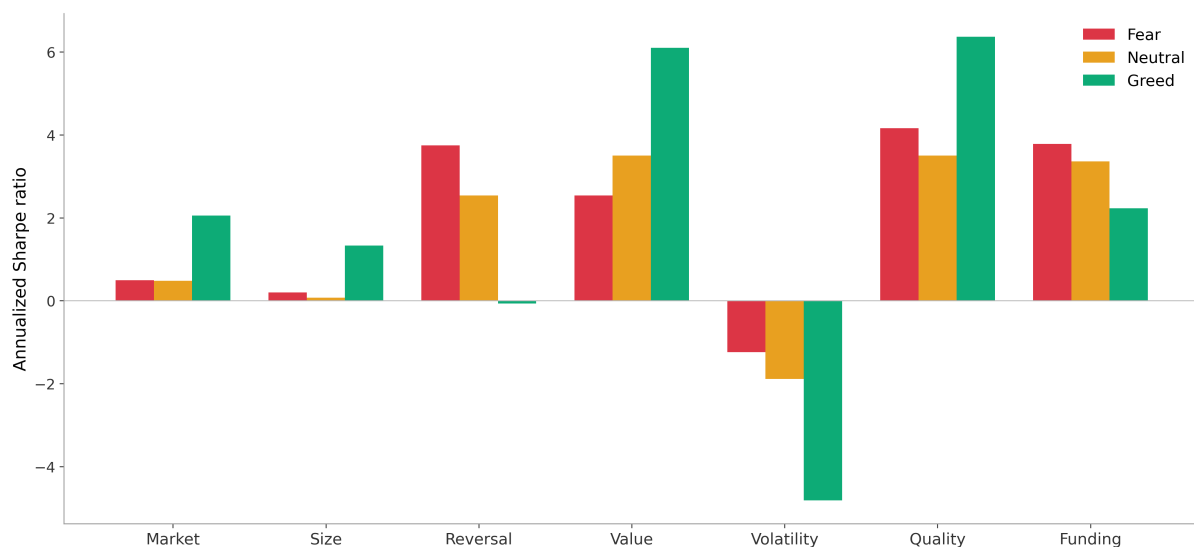
Quality is the only factor that maintains strong performance across all regimes, with annualized returns of +184% in fear, +178% in neutral, and +424% in greed. This regime independence strengthens the case that on-chain quality captures a genuine, persistent dimension of return variation rather than a sentiment-driven artifact.

Funding maintains positive returns in all environments, with its strongest performance during fear (+140%) when crowded short positions are most prone to violent squeezes.

**Table 12:** Factor Premia by Sentiment Regime

| Factor   | Fear      |       | Neutral   |       | Greed     |       |
|----------|-----------|-------|-----------|-------|-----------|-------|
|          | Ann. Ret. | SR    | Ann. Ret. | SR    | Ann. Ret. | SR    |
| REV      | +234.6%   | +3.75 | +180.3%   | +2.54 | −5.1%     | −0.07 |
| VAL      | +93.9%    | +2.54 | +139.6%   | +3.50 | +365.2%   | +6.10 |
| VOL      | −121.2%   | −1.24 | −182.6%   | −1.89 | −452.1%   | −4.82 |
| QUA      | +183.5%   | +4.16 | +177.6%   | +3.50 | +424.4%   | +6.37 |
| FND      | +140.0%   | +3.78 | +138.3%   | +3.36 | +128.0%   | +2.23 |
| Market   | +32.6%    | +0.49 | +29.8%    | +0.48 | +128.7%   | +2.05 |
| FM $R^2$ | 0.331     |       | 0.344     |       | 0.323     |       |
| Days     | 746       |       | 947       |       | 613       |       |

Regimes defined by the 30-day smoothed **Alternative.me Fear & Greed Index**. SR denotes annualized Sharpe ratio.



**Figure 6:** Factor Sharpe ratios by sentiment regime.



**Figure 7:** Annualized factor returns by sentiment regime.

Figures 6 and 7 make the regime dependence visually apparent. The magnitude differences are most pronounced for the reversal factor (strong in fear, flat in greed) and the value factor (moderate in fear, dominant in greed). Quality and funding maintain positive returns across all environments.

We also test whether the Fear & Greed Index predicts aggregate market returns. The correlation between the daily index value and forward market returns is positive at all horizons ( $r = +0.16$  at 63 days,  $t = 7.52$ ), indicating that sentiment exhibits momentum rather than mean reversion at the market level. The popular contrarian interpretation of the index does not hold in this sample, though the predictive  $R^2$  remains small ( $<3\%$ ).

## 7 Limitations

Several limitations warrant discussion.

**Survivorship bias inflates all reported factor returns.** Our universe consists of tokens alive and liquid as of May 2026. Projects that failed, were delisted, or lost 99% of their value are absent. The tokens that “survived” are disproportionately those with high network quality and sufficient market capitalization, precisely the characteristics our quality and value factors reward.

**On-chain coverage is partial.** The value and quality factors are available for approximately 52 to 53 of 90 tokens. The remaining tokens receive neutral (zero) exposure in cross-sectional regressions. If the missing tokens systematically differ from the covered ones (which is likely, since coverage correlates with blockchain indexing priority), the estimated factor premia may be biased.

**Factor return magnitudes are not achievable.** The reported annualized returns (e.g., quality at +245%) reflect quintile long-short spreads computed without transaction costs, market impact, or capacity constraints. Cryptocurrency bid-ask spreads, particularly for smaller tokens, can exceed 1% per trade. Real-world implementation would produce substantially lower returns.

**The size factor is insignificant.** After orthogonalizing volatility to size, the size premium disappears ( $t = 1.36$ ). We retain size in the model because it serves as an orthogonalization control for the value, quality, and funding factors, and because its incremental time-series  $R^2$  contribution (2.2%) suggests it captures some variation even if it is not independently priced in the cross-section.

**The reversal finding contradicts [Liu et al. \(2022\)](#).** This discrepancy may reflect differences in sample composition (90 tokens versus their larger universe), sample period (our data extends four years beyond theirs), or data source quality. Further investigation of the conditions under which momentum versus reversal dominates in crypto markets would be valuable.

## 8 Conclusion

The central message of this paper is that cryptocurrency returns are not random. They are driven by measurable token characteristics, some familiar from equity markets, others native to blockchain networks, and these characteristics can be organized into a coherent factor model with essentially zero overfitting.

The Q-7 framework extends the [Liu et al. \(2022\)](#) three-factor model with fundamental valuation, residual volatility, on-chain quality, and derivatives sentiment, while re-

placing momentum with short-term reversal. Using a single institutional data source for 90 digital assets over six years, five of six style factors carry statistically significant risk premia at the 1% level.

Four findings merit emphasis. First, **on-chain network quality is the single most reliable predictor of cryptocurrency return differences**. The quality factor maintains statistical significance across all sentiment regimes, suggesting it captures a fundamental dimension of return variation that is not driven by speculative dynamics. As blockchain data coverage improves, this signal should become exploitable for a broader set of tokens.

Second, **short-term reversal, not momentum, drives cross-sectional return predictability**. Past losers outperform past winners at the 21-day horizon, contradicting the momentum effect documented by [Liu et al. \(2022\)](#). This reversal is strongest during fearful markets and operates within sectors, not across them.

Third, **the apparent size premium in cryptocurrency markets is a volatility premium in disguise**. After orthogonalizing volatility to size, the residual size effect vanishes. Small tokens outperform not because they are small, but because they are volatile, and bearing idiosyncratic volatility is compensated in a market dominated by speculative retail participation.

Fourth, **factor premia are regime-dependent in economically intuitive ways**. Reversal is a fear-regime phenomenon: panic selling creates overshooting that subsequently corrects. Value is a greed-regime phenomenon: optimistic capital flows discriminate between fundamentally cheap and expensive tokens. Quality transcends regimes entirely. Practitioners who condition their factor tilts on prevailing sentiment may be able to harvest premia more efficiently, though this conditioning analysis is in-sample and would benefit from further validation.

The Q-7 model is designed for portfolio risk decomposition and performance attribution. Its primary use case is answering a practical question: when a digital asset portfolio goes up or down, how much is attributable to the market, to sector allocation, and to specific factor exposures? For institutional allocators entering the digital

asset space, this decomposition is table stakes.

Future work should address the survivorship bias limitation through point-in-time universe construction, extend on-chain data coverage to more tokens, and investigate whether the reversal effect is robust to alternative data sources and time periods.

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